LINUX PROGRAMMING : ASSIGNMENT-4

1. A system has a file /etc/passwd. How would you use grep + tee to extract usernames and save them to a file while also displaying them on screen?

ANS The /etc/passwd file contains user records separated by colons. Usernames are the first field.
Command Pipeline:
grep -v "^#" /etc/passwd | cut -d: -f1 | tee extracted_usernames.txt

1. [cite_start]grep -v "^#": Filters out lines starting with the comment character (#).
2. [cite_start]cut -d: -f1: Uses the cut command to select (-f1) the first field, using the colon (:) as the delimiter (-d).
3. [cite_start]tee extracted_usernames.txt: Takes the resulting list of usernames and directs it to the terminal and simultaneously saves it to the file.

2. A binary isn't found in $PATH. How would you use commands (which, find, locate) to troubleshoot and fix the issue?

ANS   Troubleshooting involves confirming the binary's absence from the search path, locating its actual position, and then fixing the environment variable.
1. Initial Check and Diagnostic:
* which binary_name: Confirms the shell cannot find the binary in any directory listed in the current $PATH.
* echo $PATH: Displays the existing search path directories.
2. Locating the Binary:
* [cite_start]locate binary_name: Use this first, as it is fast, to check the database for the file's location.
* [cite_start]find / -name binary_name 2>/dev/null: If locate fails, perform a thorough, real-time filesystem search starting from the root directory (/).
3. Fixing the $PATH:
* Once the absolute directory is found (e.g., /opt/custom/bin), the directory must be added to the $PATH variable.
* Temporary Fix: export PATH=$PATH:/opt/custom/bin
* Permanent Fix: Add the export line to the user's shell configuration file (e.g., ~/.bashrc) and reload the configuration (source ~/.bashrc).

3. Write a command pipeline that finds all .log files modified in the last 24 hours in /var/log and saves results into log_report.txt

ANS The pipeline uses the find command with time criteria and redirects the final output.
Command

find /var/log -type f -name "*.log" -mtime -1 > log_report.txt

* [cite_start]find /var/log: Starts the search in the log directory.
* -type f: Ensures only regular files are included.
* [cite_start]-name "*.log": Matches files ending with the .log extension.
* [cite_start]-mtime -1: Filters for files modified less than one full day (24 hours) ago.
* > log_report.txt: Directs the list of files to the specified file, overwriting existing content.

4. What is the difference between shutdown -r now and reboot?

ANS • shutdown -r now
- Gracefully stops all running processes and services before restarting.
- Notifies logged-in users about the shutdown/restart event.
- Writes shutdown logs to system files.
- Gives administrators more control (e.g., scheduling shutdowns, sending custom messages).

- reboot
- Directly calls the system reboot function.
- Usually faster, but less graceful — may not notify users or stop services properly (depends on system configuration).
- Best used for quick restarts, not for planned/scheduled ones.

5. How can you use the tee command to debug a script that generates both standard output and error messages?

ANS Pavan: To capture both standard output (stdout, file descriptor 1) and standard error (stderr, file descriptor 2) simultaneously, the stderr stream must first be redirected into the stdout stream before piping to tee.
Pavan: ./buggy_script.sh 2>&1 | tee script_trace.log
[Pavan: 2>&1: This critical part redirects file descriptor 2 (stderr) to file descriptor 1 (stdout).
* [cite_start]| tee script_trace.log: The combined stream is then piped to tee, which prints all output (regular and errors) to the terminal for live debugging and saves it to script_trace.log

6. Explain any three real-world applications of Linux in industries.

ANS Enterprise Servers and Cloud Infrastructure: Linux powers the majority of web servers (e.g., Apache, Nginx) and is the dominant operating system for cloud platforms (AWS, Azure, Google Cloud). Its ability to handle massive workloads and its low resource usage make it ideal for data centers
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2. Supercomputing and Scientific Research: Nearly all of the world's top supercomputers run on Linux. Its open and highly configurable kernel allows researchers to optimize it for parallel processing tasks necessary for molecular modeling, physics simulations, and weather forecasting.

3. Android and Embedded Devices: The Android mobile operating system is built on the Linux kernel. [cite_start]Furthermore, custom Linux variants are deeply integrated into

embedded systems like smart TVs, digital video recorders (DVRs), routers, and industrial control systems due to their stability and low cost.

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7. Differentiate application, system and utility software in the context of Linux environment.

ANS •Application Software

- Programs designed for end-users to perform specific tasks.
- Examples: LibreOffice (document editing), Firefox (web browsing), GIMP
- Runs on top of system software and depends on it to function.
- System Software
- Provides the core functions that make the computer usable.
- Includes the Linux kernel, device drivers, and operating system components.
- Manages hardware resources and provides a platform for applications.
- Utility Software
- Specialized tools that support system maintenance, optimization.
- Examples: top, htop (process monitoring), fsck (file system check),
- Bridges the gap between system software and users for managing tasks efficiently.

8.What are the key differences between open-source and proprietary operating systems?

ANS • Source Code Availability

- Open-source: Source code is publicly available for use, modification, and distribution.

- Proprietary: Source code is closed; only the vendor/developer has access.
- Cost
- Open-source: Usually free to use, though paid enterprise support may exist.
- Proprietary: Typically requires purchase of a license or subscription.
- Customization
- Open-source: Users can modify and tailor the OS to their needs.
- Proprietary: Customization is limited to what the vendor allows.
- Support
- Open-source: Community-based forums, documentation, and optional paid support.

- Proprietary: Official vendor support, often with guaranteed service levels.
- Security
- Open-source: Transparency allows vulnerabilities to be identified and patched quickly by the community.

- Proprietary: Security fixes depend solely on the vendor’s response time.
- Examples
- Open-source: Linux distributions (Ubuntu, Fedora), FreeBSD.

- Proprietary: Microsoft Windows, macOS

9. Write the command to display the system’s kernel version

ANS The uname command is used to output system information; the -r option specifies the kernel release version.
Command: uname -r

10. What is the difference between head and tail commands in text processing?

ANS [cite_start]These commands are used to display partial contents of a file, differing by their starting point.
* [cite_start]head: Displays content from the beginning of a file. By default, it outputs the first 10 lines.
* [cite_start]tail: Displays content from the end of a file. By default, it outputs the last 10 lines. [cite_start]It is particularly useful with the -f (follow) option for monitoring log files as they are written to.